

GomMold

Detect early, breathe easy.

SOFTWARE & AI ENGINEERING PROJECT

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INTRODUCTION

PROBLEM

International students and residents in South Korea often struggle to identify early-stage indoor mold due to unfamiliar weather conditions and subtle visual signs.

QUESTION

How can we provide a simple and reliable way for people to detect mold early and receive clear cleaning guidance ?



IDEA

Develop a smartphone app that uses AI to identify mold from photos and instantly offer practical cleaning advice.

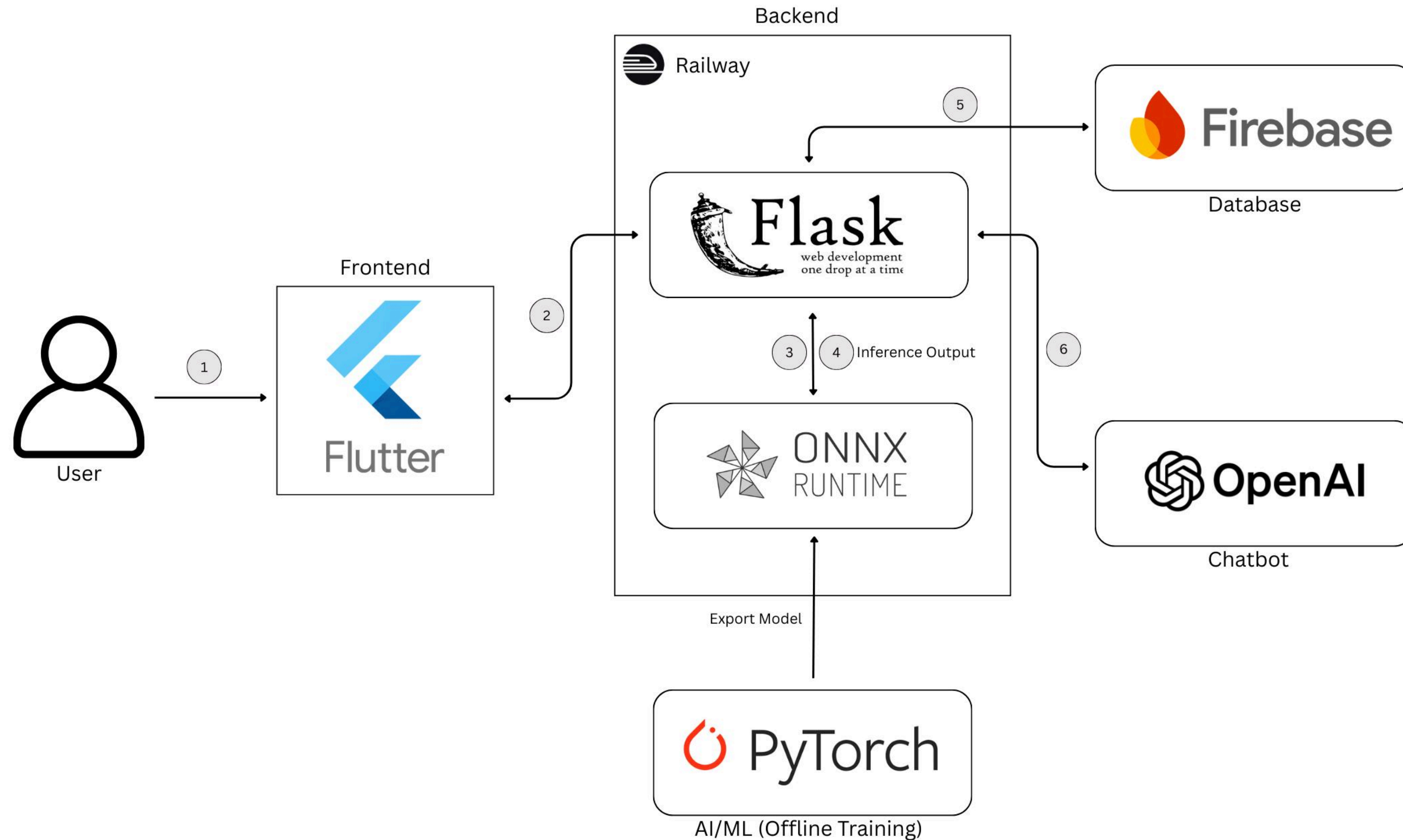
GomMold

AI Driven Mold Detection

[DEMO VIDEO](#)



ARCHITECTURE



FIRESTORE DATABASE

Users Collection

- email
- username
- password (hashed)
- auto-generated user_id

Detections Collection

1. When Mold is Detected

- result: "warning"
- message: "Mold detected"
- predictions: list of bounding boxes (x1, y1, x2, y2, confidence)
- Color code: Red
- Saved along with image URL, timestamp, and user ID

2. When No Mold is Detected

- result: "safe"
- message: "No mold detected"
- predictions: empty list []
- Color code: Green
- Image still saved, with analysis name and timestamp

BEFORE



Dataset: Mold & Clean Wall Images

Collected from multiple sources (internet, our own photos).

Then → preprocess & annotate (bounding boxes).

Process Data

Resize images, normalize them, and organize labels in YOLO format.

Result (Before Training)

Model does not recognize mold.

AFTER



Train

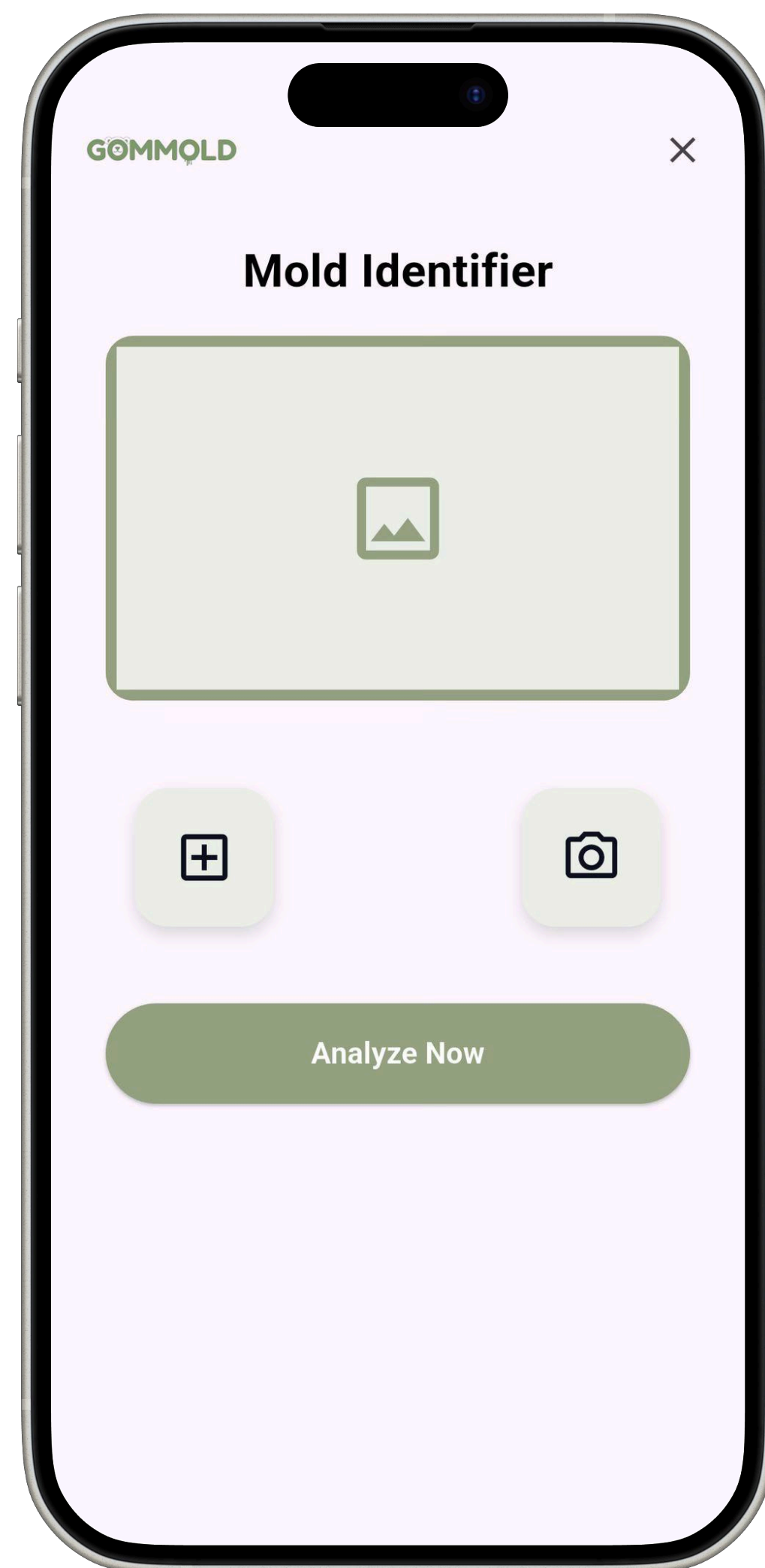
Compute loss per batch → backpropagation updates YOLO weights.

Convert

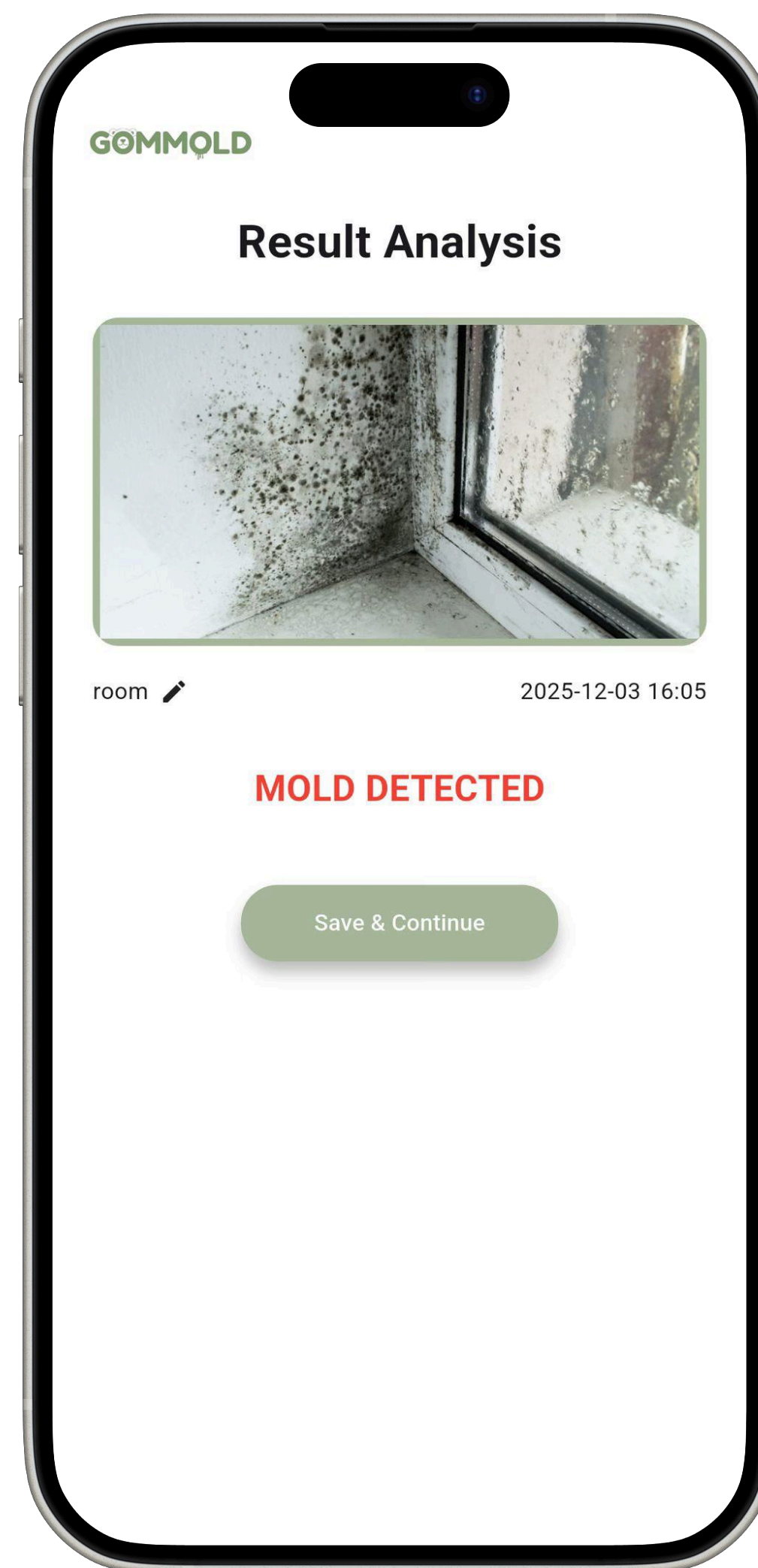
After training, export PyTorch model → ONNX for cloud deployment.

Result (After Training)

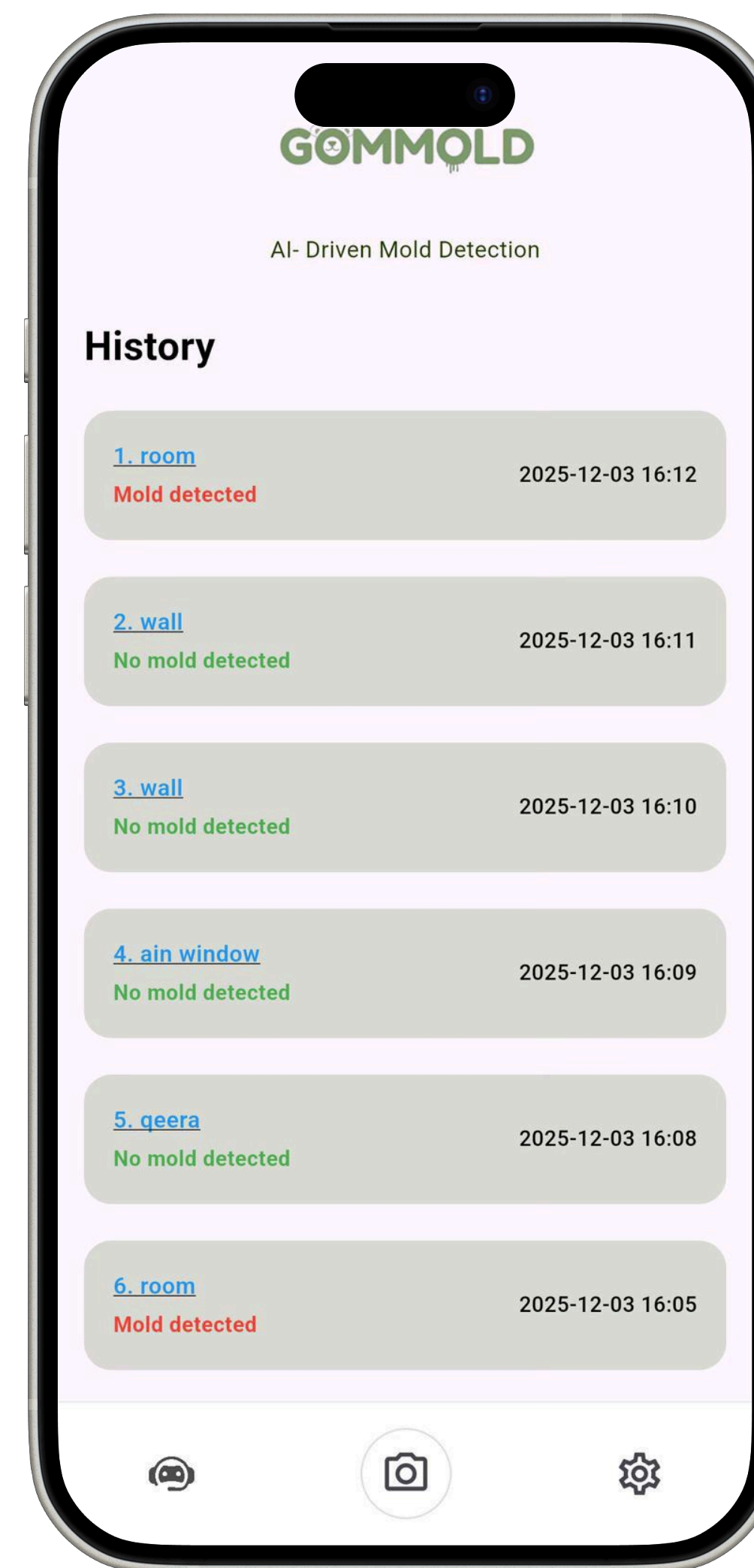
Model detects mold spots with clearer bounding boxes and higher confidence.



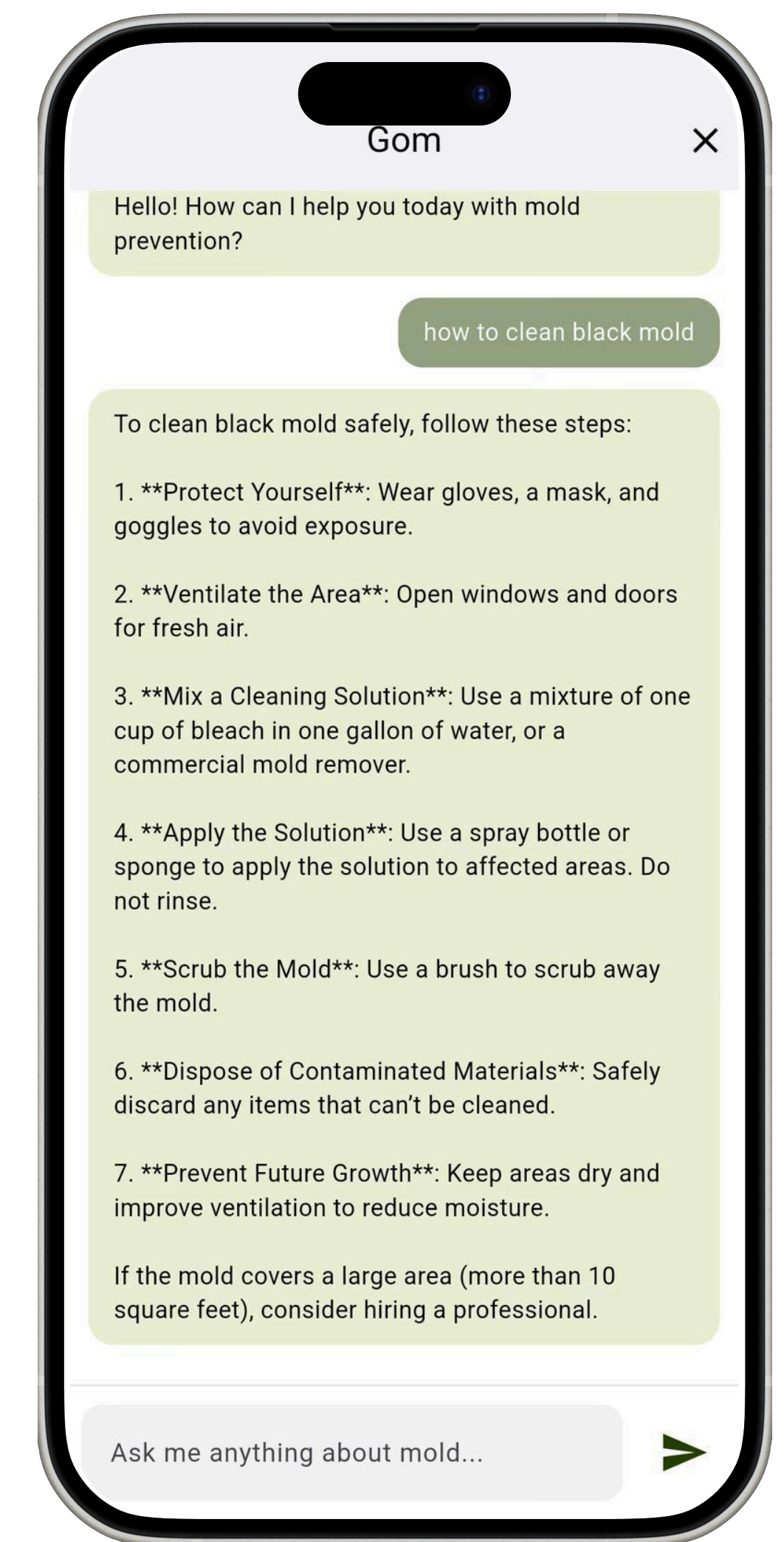
Key feature 1



Key feature 2



Key feature 3



Key feature 4



Thank You